Ganita Manjari — Grade 9 (Part II)

Chapter 14: Math of Space (Surface Area and Volume) — Exercise Set 14.4 Solutions

Question 1

A ball bearing has a radius of 0.7 cm. Find its surface area.

Easy Step-by-Step Solution:

Surface Area of Sphere = $4\pi r^2$

$SA = 4 \times (22/7) \times (0.7)^2 = 4 \times (22/7) \times 0.49 = 4 \times 22 \times 0.07 = 6.16\ cm^2$

Final Answer: 6.16 cm²

Question 2

Two solid spheres made of the same metal have weights 5920 g and 740 g. Determine the radius of the larger sphere, if the diameter of the smaller one is 5 cm.

Easy Step-by-Step Solution:

Since both spheres are made of the same metal, their mass ratio equals their volume ratio.

$V_1 / V_2 = 5920 / 740 = 8$

Volume ratio of spheres is also $(R_1 / R_2)^3 = 8$

$R_1 / R_2 = \sqrt[3]{8} = 2$

Smaller sphere diameter = 5 cm ⇒ Smaller radius $R_2 = 2.5\ cm$.

Larger radius $R_1 = 2 \times R_2 = 2 \times 2.5 = 5\ cm$.

Final Answer: 5 cm

Question 3

The diameter of the moon is approximately one fourth the diameter of the earth. Given that the moon and earth are both roughly spherical, find the ratio of their surface areas.

Easy Step-by-Step Solution:

Let diameter of Earth = d ⇒ Radius of Earth = r_E.

Diameter of Moon = $d / 4$ ⇒ Radius of Moon = $r_M = r_E / 4$.

Surface Area Ratio = $(4\pi r_M^2) / (4\pi r_E^2) = (r_M / r_E)^2 = (1/4)^2 = 1 / 16$

Final Answer: 1 : 16

Question 4

Find (i) the curved surface area and (ii) the total surface area of a hemisphere of radius 21 cm.

Easy Step-by-Step Solution:

(i) Curved Surface Area (CSA):

$CSA = 2\pi r^2 = 2 \times (22/7) \times 21^2 = 2 \times 22 \times 3 \times 21 = 2772\ cm^2$

(ii) Total Surface Area (TSA):

$TSA = 3\pi r^2 = 3 \times (22/7) \times 21^2 = 3 \times 22 \times 3 \times 21 = 4158\ cm^2$

Final Answers: (i) 2772 cm² | (ii) 4158 cm²

Question 5

Metal spheres, each of radius 2 cm, are packed into a rectangular box of internal dimensions 16 cm × 8 cm × 8 cm. When 16 spheres are packed, the box is filled with preservative liquid. Find the volume of this liquid. Round your answer to the nearest integer.

Easy Step-by-Step Solution:

Volume of box = $16 \times 8 \times 8 = 1024\ cm^3$

Volume of 1 sphere = $(4/3)\pi r^3 = (4/3) \times (22/7) \times 2^3 = (4/3) \times (22/7) \times 8 = 704 / 21 \approx 33.52\ cm^3$

Volume of 16 spheres = $16 \times (704 / 21) = 11264 / 21 \approx 536.38\ cm^3$

Volume of liquid = $\text{Volume of box} - \text{Volume of 16 spheres} = 1024 - 536.38 = 487.62\ cm^3$

Final Answer: 488 cm³

Question 6

The radius of a sphere is increased by 10%. Show that the volume increases by approximately 33.1%.

Easy Step-by-Step Solution:

Let original radius = r ⇒ Original volume $V = (4/3)\pi r^3$

New radius $r' = 1.10\ r$

New volume $V' = (4/3)\pi\ (1.10\ r)^3 = 1.331 \times [(4/3)\pi\ r^3] = 1.331\ V$

Percentage increase = $[(V' - V) / V] \times 100\% = (1.331 - 1) \times 100\% = 33.1\%$

Final Answer: Proved (33.1% increase)

Question 7

The radius of a sphere is increased by x%. The volume of the sphere increases by 72.8%. Find the value of x.

Easy Step-by-Step Solution:

New volume $V' = 1.728\ V$

Since $V' / V = (r' / r)^3 = 1.728$:

$r' / r = \sqrt[3]{1.728} = 1.2$

Percentage increase $x = (1.2 - 1) \times 100\% = 20\%$

Final Answer: x = 20%

Question 8

The hemispherical dome of a building needs to be painted. If the circumference of the base of the dome is 35.2 m, find the cost of painting it, given the cost of painting is ₹ 10 per 100 cm².

Easy Step-by-Step Solution:

Step 1: Find radius (r)

Circumference $= 2\pi r = 35.2\ m \Rightarrow 2 \times (22/7) \times r = 35.2 \Rightarrow r = (35.2 \times 7) / 44 = 5.6\ m$

Step 2: Calculate Curved Surface Area

$CSA = 2\pi r^2 = 2 \times (22/7) \times 5.6^2 = 2 \times (22/7) \times 31.36 = 197.12\ m^2$

Step 3: Convert rate and calculate total cost

Rate = ₹ 10 per 100 cm² = ₹ 10 per 0.01 m² = ₹ 1000 per m².

Total Cost = $197.12\ m^2 \times ₹\ 1000/m^2 = ₹\ 1,97,120$

Final Answer: ₹ 1,97,120